CASE D — Mr. John Harris

72-year-old Male · HFrEF — | Decompensated · Secondary to Ischaemic Heart Disease

Target: 70%+ | Physiotherapy

■ PROBLEM LIST

- Reduced cardiac output (impaired EF)
- Pulmonary oedema — impaired alveolar gas exchange
- Bilateral peripheral oedema — reduced ankle mobility
- Orthopnoea — positional dyspnoea
- Skeletal muscle deconditioning / cardiac cachexia
- Impaired peripheral oxygen delivery
- Severely reduced exercise tolerance
- Dyspnoea completing ADLs (showering, dressing)
- Sleep disruption from orthopnoea / PND
- Reduced walking distance / community mobility
- Social isolation and reduced quality of life

■ RED FLAGS

- ■ Acute pulmonary oedema (sudden severe dyspnoea, SpO■ <88–90%, pink frothy sputum)
- Immediate life threat — fluid flooding the alveoli prevents effective gas exchange. Patient may deteriorate rapidly and require NIV or emergency intubation.
- Action: Cease all physiotherapy immediately. Sit patient fully upright. Apply supplemental O■. Call medical emergency. Do not leave patient alone.
- ■ Chest pain — new or changing (possible ACS)
- Given his ischaemic heart disease aetiology, any new chest pain — particularly exertional — must be treated as potential ACS until ruled out. Exercise during active ischaemia can cause fatal arrhythmia.
- Action: Stop all activity immediately. Obtain 12-lead ECG urgently. Alert medical team. Do not resume exercise until cleared.
- ■ Significant arrhythmia (new AF, rapid ventricular response, VT, HR >120 or <40 bpm at rest)
- Arrhythmias are common in HFrEF and can cause haemodynamic compromise, especially with any physical exertion. Exercise is unsafe until rhythm is controlled.

■ PATHOPHYSIOLOGY

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■ DIFFERENTIAL DIAGNOSES

1. COPD Exacerbation Supporting features 72-year-old male — typical demographic for COPD Worsening dysp

- 72-year-old male — typical demographic for COPD
- Worsening dyspnoea and markedly reduced exercise tolerance
- CXR changes may superficially overlap (basal changes)

2. Bilateral Pleural Effusions (non-cardiac cause) Supporting features Bilateral CXR changes and progre

- Bilateral CXR changes and progressive dyspnoea
- Orthopnoea possible with large bilateral effusions
- Bilateral ankle oedema if hypoalbuminaemia or renal disease

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Resting Vital Signs + Orthostatic Blood Pressure Assessment

What it measures: Haemodynamic stability — determines whether it is safe to proceed with any physiotherapy intervention. This is the non-negotiable first step in every decompensated HF session. How to perform: Record HR, BP, SpO■, RR, and temperature at rest in supine, then sitti

1-Minute Sit-to-Stand Test (1MSTS)

What it measures: Functional exercise capacity and physiological response to exertion — allows safe, brief quantification of cardiovascular and muscular functional reserve without the haemodynamic demands of a 6MWT. How to perform: Standard chair (no armrests, approximately 45 cm

NYHA Functional Class Classification

What it measures: Standardised clinical classification of functional limitation severity in heart failure — universally recognised, enables communication with the cardiology team, and directly informs exercise prescription intensity. How to perform: Through structured subjective

■ TREATMENT PLAN